

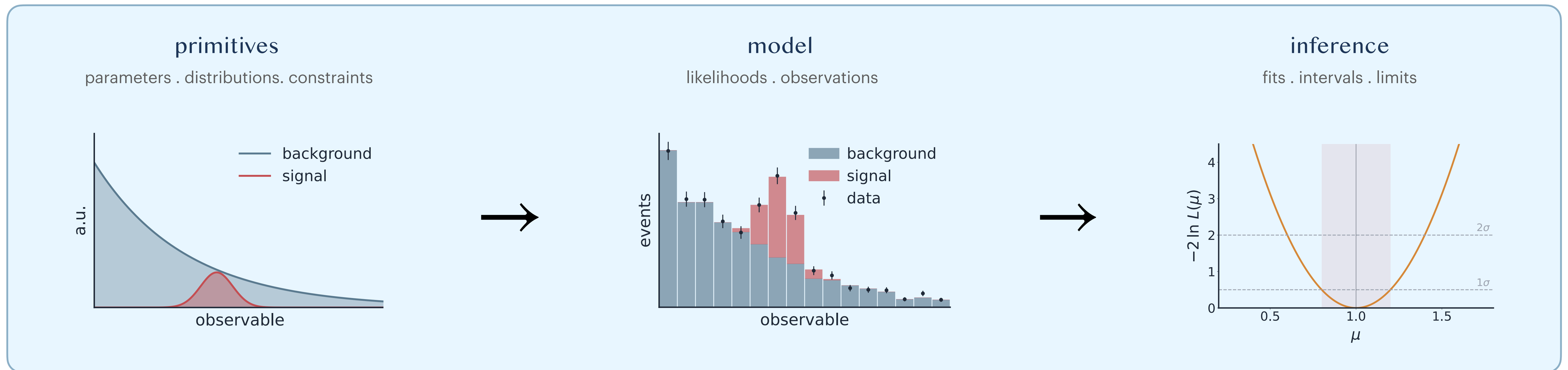


A differentiable HEP statistical analysis ecosystem with JAX

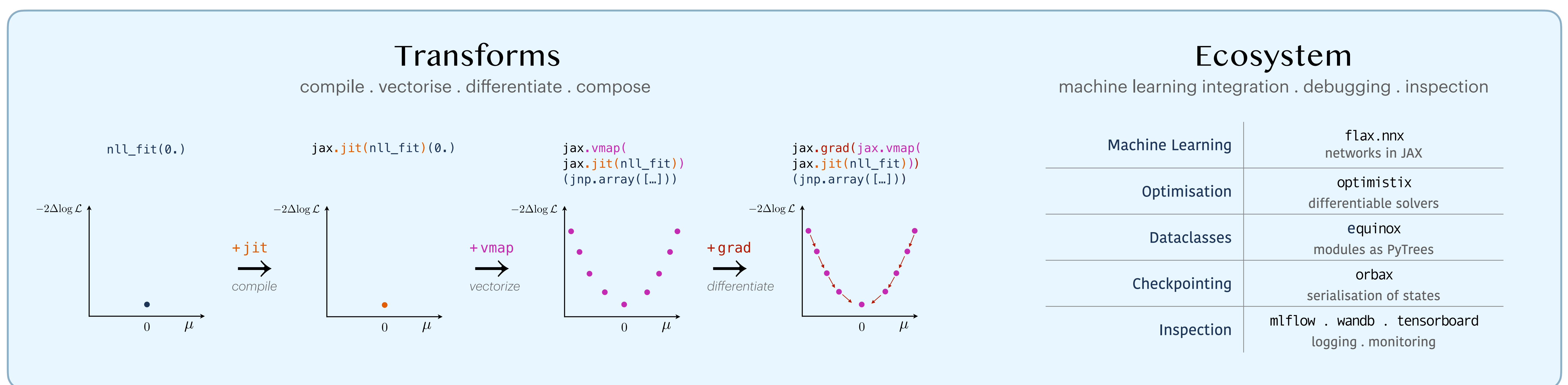


Mohamed Aly¹, Peter Fackeldey¹, Massimiliano Galli¹
Princeton University¹

Statistical Analyses in HEP



The case for JAX in HEP statistics



A fully differentiable JAX ecosystem for HEP statistics

Primitives

```
1 import jax.numpy as jnp
2 import evermore as evm
3 hist = jnp.array([...])
4 syst = evm.NormalParameter(name="syst")
5 # lnN +/- 10% on the yield
6 modifier = syst.scale_log_asymmetric(
7     up=jnp.array([1.1]),
8     down=jnp.array([0.9]),
9 )
10
11 modified_hist = modifier(hist)
```

(parameters, constraints & modifiers)

```
1 import paramore as pm
2 lower, upper = 100.0, 180.0
3 sig = pm.Gaussian(mu=125.0, sigma=2.0,
4                   lower=lower, upper=upper)
5 bkg = pm.Exponential(lambda=0.05,
6                      lower=lower, upper=upper)
7 model = pm.SumPDF(
8     pdfs=[sig, bkg],
9     extended_vals=[500.0, 5000.0],
10    lower=lower, upper=upper,
11 )
```

(distributions)

Quickstart

```
1 import paramore as pm
2 import evermore as evm
3 # parameters (a PyTree)
4 params = {
5     "sig_n": evm.Parameter(500.0),
6     "bkg_n": evm.Parameter(5000.0),
7     "mu": evm.Parameter(125.0),
8 }
9 # extended NLL from params + model
10 def nll(p, data):
11     return pm.create_extended_nll(p, model, data)
```

(unbinned model)

```
1 from jax.random import normal, key
2 import everwillow as ew
3 import everwillow.statelib as sl
4 # observed unbinned masses
5 data = normal(key(0), (10_000,)) * 2 + 125
6 # fit & parameter uncertainties
7 state = sl.State.from_pytree(params)
8 obs = {"data": data}
9
10 result = ew.fit(nll, state, obs)
11 unc = ew.uncertainties(nll, result.params, obs)
```

(likelihood-agnostic)

Interoperability and likelihood combination



Differentiable Everything!

composable . introspectable . fast . ML-friendly

Likelihood Agnostic

all you need is JAX